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LARGE LANGUAGE MODELS AS AN ENCODER

Abstract

Large language models can receive a prompt and generate responses having natural language and/or data structures as sequence of tokens. Some responses may have a few dozen tokens. The speed of response of large language models can be directly proportional to how many tokens are being generated. Rather than producing many tokens, it is possible to fine-tune a large language model to generate responses in an encoded output format. A response can have one or more encoded values that can indicate the same information as a natural language and/or structured data response. The response may include only a single or few tokens. The speed of response of a large language model operating as an encoder would be faster.

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Background/Summary

TECHNICAL FIELD

[0001] This disclosure relates generally to content item retrieval systems, and more specifically, using a large language model as an encoder in a content item retrieval system.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0002] Embodiments will be readily understood by the following detailed description in conjunction with the accompanying drawings. To facilitate this description, like reference numerals designate like structural elements. Embodiments are illustrated by way of example, and not by way of limitation, in the figures of the accompanying drawings.

[0003] Figure (FIG. 1 illustrates an implementation of a content item retrieval system, according to some embodiments of the disclosure.

[0004] FIG. 2 illustrates an implementation of context understanding part and a candidate generation part, according to some embodiments of the disclosure.

[0005] FIG. 3 illustrates an implementation of context understanding part and a candidate ranking part, according to some embodiments of the disclosure.

[0006] FIG. 4 illustrates an implementation of context understanding part that outputs one or more encoded values, and exemplary ways that the one or more encoded values may be used in a content item retrieval system, according to some embodiments of the disclosure.

[0007] FIG. 5 illustrates fine-tuning a large language model to output one or more encoded values, according to some embodiments of the disclosure.

[0008] FIG. 6 is a flowchart showing a method for content item retrieval, according to some embodiments of the disclosure.

[0009] FIG. 7 is a flowchart showing a method for ranking content items, according to some embodiments of the disclosure.

[0010] FIG. 8 is a flowchart showing a method for fine-tuning a large language model, according to some embodiments of the disclosure.

[0011] FIG. 9 is a block diagram of an exemplary computing device, according to some embodiments of the disclosure.

DETAILED DESCRIPTION

Overview

[0012] Content platforms offer users access to large libraries of content items. Users can spend a lot of time on a content platform looking for content items to consume. Finding the content items that a user is looking for can be important for user satisfaction. Also, finding the content items quickly can be important for user satisfaction. If a user is not satisfied, the user is likely not going to return to the content platform. Also, if a user is not satisfied, the user is likely not going to consume any content items.

[0013] Retrieving content items in response to a query in a way that increases user satisfaction and increases chances of users consuming a retrieved content item is not trivial. A query may include a natural language description of what a user is searching for or looking for in a library of content items. One example of a query may include, “show me some short videos about freshwater shrimp keeping”. The query from a user can be provided as input to a content item retrieval system. The content item retrieval system may return content items that are relevant to the query. Users may expect relevant search results almost immediately or instantly with very little latency or waiting time.

[0014] One way to implement a content item retrieval system is to use a self-querying retrieval technique. A large language model is used to reformulate the natural language query from the user

and output a data structure and/or a structured query. The data structure and/or the structured query can have attributes about the natural language query. The self-querying retrieval technique can be advantageous because the large language model can understand nuances and contextual cues of the natural language query and produce a context-aware data structure and/or a structured query.

Exemplary implementations of the content item retrieval system are illustrated in FIGS. 1-4.

[0015] During operation, a large language model can receive a prompt having the natural language query. The large language model may be instructed in the prompt to generate responses having natural language and/or data structures as sequence of tokens. In one example, the large language model may be instructed in the prompt to generate a structured query based on the natural language query. In another example, the large language model may be instructed in the prompt to generate a data structure having attribute-value pairs based on the natural language query. Responses that have natural language and/or data structures may have a few dozen tokens or many more. The speed of response of large language models can be directly proportional to how many tokens are being generated. If responses have many tokens, it can take much more time for the large language model to generate a response, which in turn adds latency to the process of retrieving content items relevant to the natural query back to the user.

[0016] Rather than having the large language model produce many tokens, it is possible to fine-tune the large language model to generate responses in an encoded output format. In other words, a large language model may be fine-tuned to operate as an encoder. A response generated by the large language model operating as an encoder can have one or more encoded values that can indicate the same information as a natural language and/or structured data response, but with far fewer tokens. Leveraging the large language model's ability to understand nuances and contextual cues of the natural language query, the one or more encoded values can correspond to or indicate one or more attributes about the natural language query from the user.

[0017] Advantageously, the response in an encoded output format may include only a single, a couple, three, or just a few tokens. The one or more encoded values can be efficiently translated using simple logic operations into a structured query and/or parameters, which can be used in configuring the content item retrieval system. In some cases, the one or more encoded values may be used directly as control signals to configure the content item retrieval system. The speed of response of a large language model operating as an encoder and the speed of the logic operations can be still much faster than a large language model that is generating many tokens to form a natural language response or a response having a structured query. By using the large language model operating as an encoder in a content item retrieval system, relevant content items to the natural language query may be returned to the user with lower latency.

[0018] Rather than responding to a prompt using natural language or attribute-value pairs, a large language model operating as an encoder may be prompted with an instruction having one or more questions about the natural language query from the user, each question having two or more possible answers. For example, the instruction may include a first question asking a true or false question about the natural language query, and a second question asking a multiple-choice question about the natural language query. The large language model operating as an encoder may respond to the questions using one or more encoded values that indicates a selection of an answer from the two or more possible answers. The large language model may respond to multiple questions in the form of a binary string, e.g., "01". The large language model may respond to multiple questions in the form of a cipher or symbol, e.g., "B" that corresponds to a particular combination of answers to the questions.

[0019] The one or more encoded values can be used as one or more control signals for configuring candidate generation in a content item retrieval system. The one or more control signals can configure parameters for obtaining relevant content items. The one or more control signals can be used by decoding logic to generate a structured query for obtaining relevant content items. Some detailed examples are illustrated in FIG. 4. Exemplary methods for using the one or more encoded

values to control or configure candidate generation are described in FIG. 6.

[0020] The one or more encoded values can be used as one or more control signals for configuring candidate ranking in a content item retrieval system. The one or more control signals can configure ranking parameters for ranking relevant content items. The one or more control signals can be used by decoding logic to generate ranking parameters for obtaining relevant content items. Some detailed examples are illustrated in FIG. 4. Exemplary methods for using the one or more encoded values to control or configure candidate ranking are described in FIG. 7.

[0021] Some techniques can be implemented to generate training data which can be used to fine-tune the large language model to output responses in an encoded output format. The techniques are described in FIG. 5. Exemplary methods for fine-tuning a large language model to output responses in an encoded output format are described in FIG. 8.

[0022] While the large language model as an encoder is described in the context of content item retrieval herein, it is envisioned by the disclosure that the large language model as an encoder may be implemented in other systems, e.g., in systems where extraction of attribute(s) based on a prompt may be made quicker by instructing the large language model to generate responses in an encoded output form. As long as attributes can be encoded according to an encoding scheme (and decoded appropriately based on the encoding scheme to resolve the attributes encoded in the response), and a fine-tuning process is implemented to train a large language model to operate as an encoder, the encoding scheme can be provided in the prompt to trigger the large language model to generate responses that have far fewer tokens while conveying the same information.

Semantic and Contextual Search

[0023] Content providers may manage and allow users to access and view thousands to millions or more content items. Content items may include media content, such as audio content, video content, image content, augmented reality content, virtual reality content, mixed reality content, game, textual content, interactive content, etc. Finding exactly what a user is looking for, or finding what the user may find most relevant can greatly improve the user experience. Also, finding relevant content items instantly can also improve the user experience.

[0024] FIG. 1 illustrates an implementation of a content item retrieval system **100**, according to some embodiments of the disclosure. Context **180** may be provided as input to content item retrieval system **100**. Content item retrieval system **100** may include several stages of operations. Content item retrieval system **100** may include one or more of: context understanding part **120**, candidate generation part **130**, and candidate ranking part **140**. Content item retrieval system **100** may generate results **106**.

[0025] Context **180** may capture context of a particular search or recommendation session with a user. Context **180** may capture information that may be helpful for understanding what a user is looking for and/or what may be relevant or useful to the user. In some cases, context **180** may include query **102**.

[0026] Query **102** may include natural language text and/or description provided by a user. Query **102** may include a natural language query. In some cases, query **102** may include a user-provided voice-based or text-based query to find content items. Examples of query **102** may include: [0027] “Show me funny office comedies with romance” [0028] “TV series with strong female characters” [0029] “I want to watch 1980s romantic movies with a happy ending” [0030] “Short animated film that talks about family values” [0031] “Are there blockbuster movies from 1990s that involves a tragedy?” [0032] “What is that movie where there is a Samoan warrior and a girl going on a sea adventure?” [0033] “What are some most critically-acclaimed dramas right now?” [0034] “I want to see a film set in Tuscany but is not dubbed in English” and [0035] “Recommend me movies of Brad Pitt that are free for me to watch”

[0036] In some cases, context **180** may include query **102** and optionally one or more contextual factors **170**. Examples of contextual factors **170** can include: characteristic(s) about the user making the query, time of day, day of the week, time of the year, seasonality (e.g., seasons, special

events, holidays, etc.), one or more past queries made by the user, one or more past user interactivity information with the content platform (e.g., what the user clicked on, etc.), whether the query is voice-based or text-based, the type of device that the user is using (e.g., mobile device versus television), the type of application that the user is using, whether the user is a paid subscriber or not, what subscriptions the user has, demographics about the user, whether the user is an expert/experienced user or not, whether the user is a loyal user or not, how many retrieved content items the user is looking for, characteristic(s) about the device the user is using to input the natural language query, the amount of bandwidth the user has on a network to receive content, etc. [0037] Context **180** may be provided as input to context understanding part **120**. Context understanding part **120** may process context **180** to understand context **180**, e.g., to extract contextual cues, semantic meaning, user intent, etc. In some cases, context understanding part **120** may implement a large language model. A prompt may be generated based on context **180**, and the prompt may be used as input to the large language model. Context understanding part **120** may process context **180** (e.g., receive a prompt that has information about context **180** and an instruction having questions about context **180**) and extract one or more attributes or other suitable information about context **180**.

[0038] The one or more extracted attributes or other suitable information may be provided to candidate generation part **130** to find semantically and/or contextually relevant candidates, e.g., content items semantically and/or contextually relevant to context **180**. Candidate generation part **130** may find candidates that are semantically and/or contextually relevant to context **180**.

[0039] The one or more extracted attributes or other suitable information may be provided to candidate ranking part **140** to augment ranking of relevant candidates, e.g., content items relevant to context **180**. Candidate ranking part **140** may determine a score for each relevant candidate found by candidate generation part **130** and sort the relevant candidates based on the scores to produce ranked relevant candidates.

[0040] In some cases, context understanding part **120** may generate one or more control signals based on context **180**, or output information about context **180**.

[0041] The one or more control signals and/or information may be provided to candidate generation part **130** via path **124**. In some cases, the one or more control signals and/or information may control or configure candidate generation part **130**. Details about how context understanding part **120** may control or configure candidate generation part **130** are described with FIGS. 2 and 4.

[0042] The one or more control signals and/or information may be provided to candidate ranking part **140** via path **126**. In some cases, the one or more control signals and/or information may control or configure candidate ranking part **140**. Details about how context understanding part **120** may control or configure candidate ranking part **140** are described with FIGS. 3 and 4.

[0043] Based on information from context understanding part **120**, candidate generation part **130** may determine and output relevant candidates to context **180**. Relevant candidates may be provided to candidate ranking part **140** for ranking. Candidate ranking part **140** may determine and output ranked candidates.

[0044] Content item retrieval system **100** may return results **106** having ranked relevant candidates, e.g., content items relevant to context **180**. Results **106** may be returned to the user who provided or input query **102**. Results **106** may be output (e.g., rendered for display) to the user. Results **106** may be output to the user according to the ranking determined in candidate ranking part **140**.

Self-Querying Technique: Extracting Attributes Using a Large Language Model and Applying the Attributes in Candidate Generation

[0045] FIG. 2 illustrates an implementation of context understanding part **120** and a candidate generation part **130**, according to some embodiments of the disclosure. As discussed previously context understanding part **120** may be implemented to understand the context (e.g., context **180** of FIG. 1). The understanding of the context may be used to generate a structured query **292** which may be used by candidate generation part **130** to generate candidates. Candidate generation part

130 may include one or more retrieval engines **230**. Candidate generation part **130** may include one or more content libraries **240**. The structured query **292** can be applied to one or more content libraries **240** using one or more retrieval engines **230** to retrieve relevant candidates from the one or more content libraries **240**.

[0046] One or more retrieval engines **230** may include retrieval engines employing different retrieval algorithms or strategies. One or more retrieval engines **230** may include R number of retrieval engines that may employ different retrieval strategies, e.g., retrieval engine **1 232**, retrieval engine **2 234**, . . . , and retrieval engine **R 236**. One example of a retrieval strategy is lexical match. In lexical match search, the query may be processed to extract keywords, and the keywords may be lexicographically matched against a database of content items and associated keywords. Content items which may have the greatest number of keyword lexicographic matches may be returned in response to the query. Another example of a retrieval strategy is semantic retrieval. Semantic retrieval may utilize a model to interpret the semantic meaning or context of a query and find content items that may match with the query. A model may implement natural language processing to interpret the query. A model may involve neural networks (e.g., transformer-based neural networks). A model may include a large language model. Yet another example of a retrieval strategy is a graph embedding based approach to content item retrieval. A graph embedding based approach may find a subgraph of a graph of content items which may be engaging to the user for a given query. In some cases, the graph may model relationships between content items. In some cases, the graph embedding based approach may utilize the graph to identify content items which may not be directly connected to an initial set of content items that matches the query. Yet another example of a retrieval strategy may involve returning a fixed set or list of results for a particular query. The set or list of results may be curated by editor(s), hardcoded, or predetermined. For example, a query for “presidential debate” may retrieve predetermined content items which are tapings of the most recent presidential debates, and not content items related to presidential inaugurations or state of the union addresses. Yet another example of a retrieval strategy may involve searching for content items based on user query history and/or user interactivity history information. For example, content items may be retrieved based on whether the user has launched a particular content item in the past. Yet another example of a retrieval strategy may involve searching for content items based on user profile or user characteristic(s). For example, content items may be retrieved based on demographic information about the user. Yet another example of a retrieval strategy may involve collaborative filtering. Content items may be retrieved based on interactivity with the content platform and characteristics about various users on the system. For example, content items may be retrieved based on content items viewed by users who may be similar to the current user making the query. Users may be similar to the current user if the users behaved similarly on the content platform. Users may be similar to the current user if the users are socially connected with the current user. Yet another example of a retrieval strategy may involve returning a number of content items from each clusters or buckets of content items. For example, content items may be clustered based on type or verticals (e.g., music, book, short videos, long videos, audio-only, live content, games, etc.), and a certain number of content items from each type may be returned as retrieved content items to diversify the types of content items being retrieved. The retrieved content items may have a balance of different types of content items.

[0047] One or more content libraries **240** may include libraries having different collections of content items. Candidate ranking part **140** may include C number of content libraries, e.g., content library **1 242**, content library **2, 244**, . . . , and content library **C 246**. Examples of content libraries may include: content items available with a particular subscription, free content items, free content items with advertisements, content items of a certain type (e.g., audio, video, podcasts, shorts, books, games, etc.), content items of a certain genre (e.g., thriller, sci-fi, comedy, drama, documentary, etc.), content items which are popular, content items which are not popular, content items belonging to a particular time period, content items available for purchase, content items

having a certain seasonality, content items of a certain length, content items in a particular language, content items originating from a particular region or country, content items owned by a particular distributor, content items having a certain parental rating, etc.

[0048] In some embodiments, generating a structured query **292** may include a two-stage process. The illustration in FIG. 2 illustrates an exemplary implementation of a two-stage process.

[0049] In a first stage of the two-stage process, one or more first large language models or suitable natural language processing models, e.g., in context understanding part **120**, may be used to extract one or more attributes about the context and produce a data structure, e.g., data structure **280**, that has the one or more attributes. Context understanding part **120** may provide data structure **280** to structured query builder **220** via path **124**. One or more attributes may be associated with an intent of the user or the context. One or more attributes may be associated with a regular expression that appeared in the context. One or more attributes may correspond to a retrieval engine in one or more retrieval engines **230**. One or more attributes may correspond to a content library in one or more content libraries **240**.

[0050] Data structure **280** may include attribute-value pairs. Data structure **280** may include an array of attributes (e.g., values corresponding to attributes). Data structure **280** may include one or more intents **202**. Data structure **280** may include one or more regular expressions **206**. Data structure **280** may include a selection (or indication) of one or more retrieval engine(s) **208**. Data structure **280** may include a selection (or indication) of one or more content libraries **210**.

[0051] The one or more attributes may include one or more intents **202** about the user and/or the context. A large language model may be particularly useful for extracting one or more intents **202** due to the large language model's ability to understanding contextual cues and nuances. One or more intents **202** may indicate what kind, type, or category of content items the user is searching for. One or more intents **202** may indicate one or more filters to be used in candidate generation part **130**. Examples of one or more intents **202** may include: [0052] Does the user intend to watch free content? [0053] Does the user intend to watch content that requires an active subscription? [0054] Does the user intend to watch content that can be purchased? [0055] Does the user intend to watch popular content or unpopular content? [0056] Does the user intend to watch content that is new to the user?

[0057] The one or more attributes may include one or more regular expressions **206**. One or more regular expressions **206** may include text or one or more strings, e.g., a name, a label, a tag, a genre, a title, a word, a keyword, a synopsis, a date, a year, a phrase, etc. A natural language processing model may be particularly useful for extracting one or more regular expressions **206** from the context. A large language model may also be useful for extracting one or more regular expressions **206** from the context. One or more intents **202** may indicate certain text or character patterns to be used in candidate generation part **130**.

[0058] The one or more attributes may include a selection of one or more retrieval engine(s) **208** that is determined based on the context. The context may suggest one or more ones of one or more retrieval engines **230** may be more suitable or best for the context. Selection of one or more retrieval engine(s) **208** may indicate which retrieval engine to use in candidate generation part **130**. For example, a query having, "Can you suggest a scary movie to me that I'd like because I just watched 'World's Scariest Black Cat'?" may lead to a response having one or more attributes that indicate one or more retrieval engines in one or more retrieval engines **230** may be more suitable over others (e.g., a retrieval engine employing collaborative filtering, a retrieval engine employing searching for content items based on user history/interactivity on the content platform).

[0059] The one or more attributes may include a selection of one or more content libraries **210** based on the context. The context may suggest one or more ones of one or more content libraries **240** may have specific content items that the user is searching for. Selection of one or more content libraries **210** may indicate in which one or more content libraries to look for content items in candidate generation part **130**. For example, a query having, "What's a good French-language PG-

13-rated movie available for free without advertisements?” may result in a response having one or more attributes that indicate one or more content libraries in one or more content libraries **240** may be more suitable over others (e.g., content library having free content items, content library having PG-13-rated movies, content library having French-language content items).

[0060] In some cases, selection of one or more retrieval engine(s) **208** and/or selection of one or more content libraries **210** may be derived from one or more intents **202**.

[0061] To extract one or more attributes from the context, a large language model in the first stage may receive a prompt that instructs the large language model to respond by generating a data structure that has the one or more attributes, such as data structure **280**. An example of a prompt may include:

TABLE-US-00001 PROMPT Respond as a JavaScript Object Notation data structure with following fields 1.free_for_me 2.new_to_me 3.subscription 4. popular for the query: Recommend science fiction films available for free streaming that I haven't seen yet. free_for_me denote whether user is interested in free content new_to_me denote whether user is looking for new content subscription denote whether user is interested in content that the user already has a subscription for popular denote whether user is interested in popular content

[0062] An example of a response having a data structure may include:

TABLE-US-00002 RESPONSE {free_for_me true, new_to_me true, subscription false, popular false}

[0063] The example above may include four attribute-value pairs, “free_for_me true”, “new_to_me true”, “subscription false”, “popular false”. The four fields may be the attributes. The true/false values may be the corresponding values to the attributes.

[0064] Candidate generation part **130** and/or context understanding part **120** may include structured query builder **220** to produce a structured query **292** based on the understanding of the context, e.g., the attributes about the context. In some cases, in the first stage, context understanding part **120** may output one or more attributes about the context in the form of data structure **280** and provide the attributes (e.g., via path **124**) in the form of data structure **280** to structured query builder **220**. The attributes may include information to build structured query **292**. Structured query builder **220** may produce structured query **292** based on data structure **280**.

[0065] In a second stage of the two-stage process, a second large language model or logic may be used, e.g., in data structure to structured query translator **290** of structured query builder **220** to translate the data structure **280** into structured query **292** that can be applied to one or more content libraries **240** using one or more retrieval engines **230** to retrieve relevant candidates from the one or more content libraries **240**.

[0066] The structured query **292** may follow conventions and/or syntax (or examples thereof) that is accepted or required by one or more retrieval engines **230**. Structured query **292** may include coded instructions/parameters to filter content items based on one or more intents **202**. Structured query **292** may include coded instructions/parameters to find matching content items based on one or more regular expressions **206**. Structured query **292** may include coded instructions/parameters to select, turn on, or use, one or more selected ones of one or more retrieval engines **230**. Structured query **292** may include coded instructions/parameters to disable or turn off one or more selected ones of one or more retrieval engines **230**. Structured query **292** may include coded instructions/parameters to select or use one or more selected ones of one or more content libraries **240**. A large language model in the second stage may receive a prompt that instructs the large language model to respond by generating a structured query, such as structured query **292**, that follows a certain convention/syntax and that matches the data structure, such as data structure **280**.

[0067] An example of the structured query **292** may include:

TABLE-US-00003 StructuredQuery(query='taxi driver', filter=Operation(operator =< Operator.AND: 'and'>, arguments=[Comparison (comparator =< Comparator.EQ: 'eq'>, attribute='genre', value='science fiction'), Operation(operator=<Operator.AND: 'and'>

arguments=[Comparison(comparator=<Comparator.GTE: 'gte'>, attribute='year', value=1990), Comparison(comparator=<Comparator.LT: 'lt'>, attribute='year', value=2000)], Comparison(comparator=<Comparator.EQ: 'eq'>, attribute='director', value='Luc Besson'))], limit=None)

[0068] In some embodiments, generating a structured query may include a one-stage process. A large language model may be used, e.g., in context understanding part **120**, to directly produce a structured query, e.g., structured query **292**, that can be applied to one or more content libraries **240** using one or more retrieval engines **230** to retrieve relevant candidates from the one or more content libraries **240**.

[0069] Both the two-stage process and the one-stage process can suffer from long wait times due to the number of tokens involved in the responses having a data structure and/or a structured query. Contextual Ranking Technique: Extracting Attributes Using a Large Language Model And Applying the Attributes in Candidate Ranking

[0070] FIG. **3** illustrates an implementation of context understanding part **120** and a candidate ranking part **140**, according to some embodiments of the disclosure. As discussed previously context understanding part **120** may be implemented to understand the context (e.g., context **180** of FIG. **1**). Candidate ranking part **140** may include ranking parameter(s) processing part **302**. Candidate ranking part **140** may include ranking engine **360**. The understanding of the context from context understanding part **120** may be provided to ranking parameter(s) processing part **302**. The understanding of the context may be used by ranking parameter(s) processing part **302** to generate one or more ranking parameters **366**. One or more ranking parameters **366** may be used by candidate ranking part **140** when ranking candidates. One or more ranking parameters **366** produced by ranking parameters processing part **302** may configure or augment how relevant candidates found by candidate generation part **130** (of FIGS. **1-2**) may be ranked or sorted by ranking engine **360**. Ranking engine **360** may include scoring and ranking part **310**. Ranking engine **360** may include row generation and ranking part **312**.

[0071] One or more large language models or suitable natural language processing models, e.g., in context understanding part **120**, may be used to extract one or more attributes about the context and produce a data structure, e.g., data structure **380**, that has the one or more attributes. Context understanding part **120** may provide data structure **380** to ranking parameter(s) processing part **302** via path **126**. One or more attributes may be associated with an intent of the user or the context. One or more attributes may be associated with one or more rows that may be relevant or of interest to the user or the context.

[0072] In some cases, relevant candidates may be output to a user in rows or groups of content items. A row may correspond to a subcategory or subgroup. A row may correspond to a tag, a characteristic, or a dimension about content items. Examples of rows may include: "popular", "new to you", "has strong female-leads", "1990s throwback", etc. Rows may help organize/arrange/divide relevant candidates into smaller subgroups, which may help the user browse and find content that the user is looking for more easily or quickly.

[0073] Data structure **380** may include attribute-value pairs. Data structure **380** may include an array of attributes (e.g., values corresponding to attributes). Data structure **380** may include one or more intents **202**. Data structure **380** may include one or more rows **308**.

[0074] The one or more attributes may include one or more intents **202** about the user and/or the context. Ranking parameter(s) processing part **302** may translate one or more intents **202** into one or more ranking parameters **366** usable by scoring and ranking part **310**. For example, ranking parameter(s) processing part **302** may determine that one or more intents **202** indicate the user is looking for content available with subscription. Ranking parameter(s) processing part **302** may assign a weight or boosting parameter that can make content available with subscription rank higher in the results. Ranking parameter(s) processing part **302** may produce a weight or boosting parameter and provide the weight or boosting parameter as part of one or more ranking parameters

366. Scoring and ranking part **310** may use the weight or boosting parameter as part of one or more ranking parameters **366** when assigning scores to relevant candidates. Scoring and ranking part **310** may then rank/sort the relevant candidates according to the scores.

[0075] The one or more attributes may include one or more rows **308** may be relevant or of interest to the user or the context. Ranking parameter(s) processing part **302** may translate one or more intents **202** into one or more ranking parameters **366** usable by row generation and ranking part **312**. For example, ranking parameter(s) processing part **302** may determine that one or more rows **308** indicate the user may be interested looking for content having these dimensions: “watch again”, “cult classics”, and “inspired by Mel Brooks”. Ranking parameter(s) processing part **302** may assign filters based on the one or more rows **308**. Ranking parameter(s) processing part **302** may produce the filters and provide the filters as part of one or more ranking parameters **366**. Row generation and ranking part **312** may use the filters as part of one or more ranking parameters **366** to create subgroups of relevant candidates. Row generation and ranking part **312** may determine scores for the relevant candidates. Row generation and ranking part **312** may use the filters may rank/sort the relevant candidates within each subgroup separately according to the scores.

[0076] Using a large language model to produce data structure **380** would suffer from long wait times due to the number of tokens involved in the responses having a data structure such as data structure **380**.

Addressing the Speed of Response of the Large Language Model by Modifying the Large Language Model to Produce Responses in an Encoded Output Format

[0077] Referring back to FIGS. 2-3, one or more large language models may be implemented, e.g., in context understanding part **120**, to produce one or more of: a data structure, and a structured query. These large language models may be slow due to the large language models having to generate a few dozen or more tokens. Instead of using such language models, a language model operating as an encoder may be implemented instead.

[0078] FIG. 4 illustrates an implementation of context understanding part **120** that outputs one or more encoded values, and exemplary ways that the one or more encoded values may be used in a content item retrieval system, according to some embodiments of the disclosure. Rather than passing on attributes and/or data structures to structured query builder **220** via path **124**, context understanding part **120** may output one or more encoded values **410** to structured query builder **466** via path **124**. Structured query builder **466** can include a modified implementation (when compared to structured query builder **220** of FIG. 2) to utilize one or more encoded values **410** to generate a structured query **292**. Rather than passing on attributes and/or data structures to ranking parameter(s) processing part **302** via path **126**, context understanding part **120** may output one or more encoded values **420** to ranking parameter(s) processing part **488** via path **126**. Ranking parameter(s) processing part **488** can include a modified implementation (when compared to processing part **302** of FIG. 3) to utilize one or more encoded values **410** to generate one or more ranking parameters **366**.

[0079] A content item retrieval system may implement one or more decoder functions to decode the response having one or more encoded values (e.g., one or more encoded values **410** and one or more encoded values **420**). The one or more decoder functions may determine one or more attributes to which the one or more encoded values in the response correspond. The one or more decoder functions may produce structured queries and/or ranking parameters that may be used in a component of the content item retrieval system. Even when a decoder function is implemented to decode the response, the overall speed of content retrieval is faster because the large language model does not need to generate natural language responses, data structures, or structured queries. Exemplary decoder functions are illustrated as encoded values to structured query translator **460** and encoded values to ranking parameter(s) translator **480**.

[0080] In some embodiments, context understanding part **120** may include one or more of: one or more large language models and one or more natural language processing models. The one or more

models may operate as an encoder. The one or more models may output one or more encoded values, e.g., one or more encoded values **410** and one or more encoded values **420**.

[0081] Context understanding part **120** may receive a context (e.g., context **180** of FIG. **1**). The context may include a natural language query from a user. The context may include one or more contextual factors about the user or the context. The context may include one or more contextual factors about a user session. The context may include one or more contextual factors about device used by the user.

[0082] Context understanding part **120** may input or provide, to a large language model in context understanding part **120**, a prompt comprising the context and an instruction to produce a response about the context in an encoded output format. The large language model may receive the prompt. An example of a prompt may include:

TABLE-US-00004 PROMPT Produce a binary string as output that satisfies the following 1. If user intent is to watch a free movie then 1 else 0 2. if user has mentioned an actor name in the query then 1 else 0 3. if user has mentioned a title in query then 1 else 0 Respond for the user query “recommend me movies of Brad Pitt that are free to watch”

[0083] The prompt above is instructing the large language model to respond with information about three attributes: “whether user intent is to watch a free movie”, “whether user has mentioned an actor name in the query”, and “whether user has mentioned a title in query”.

[0084] The large language model operating as an encoder may generate a response based on the prompt. The response may include one or more encoded values. The one or more encoded values may correspond to one or more attributes about the context. In response to the prompt above, the large language model operating as an encoder may generate a response having one or more encoded values that correspond to the three attributes. An example of a response may include:

TABLE-US-00005 RESPONSE 110

[0085] In the example above, the response may include just a single token, “110”. However, the single token can effectively communicate and include information about three attribute-value pairs, “whether user intent is to watch a free movie YES”, “whether user has mentioned an actor name in the query YES”, and “whether user has mentioned a title in query NO”. The single token may indicate that the user intent is to watch a free movie, that the user has mentioned an actor name, and that the user has not mentioned a title.

[0086] Depending on the implementation of the large language model, the response in the example above may include three tokens, “1”, “1”, “0”. The three tokens can effectively communicate and include information about three attribute-value pairs, “whether user intent is to watch a free movie YES”, “whether user has mentioned an actor name in the query YES”, and “whether user has mentioned a title in query NO”. The three tokens may indicate that the user intent is to watch a free movie, that the user has mentioned an actor name, and that the user has not mentioned a title.

[0087] The large language model operating as an encoder may generate a response having far fewer tokens than a large language model that is generating a natural language response, or a large language model that is generating one or more attribute-value pairs (e.g., a data structure). Therefore, the large language model operating as an encoder may operate much faster, which may in turn decrease latency of returning results back to the user.

[0088] A large language model operating as an encoder can be used effectively when the response expected from the large language model predicts attributes about the context. The large language model operating as an encoder may be instructed to respond using one or more encoded values that represent or encode one or more attributes about the context, as opposed to instructing the large language model to respond using natural language, a data structure, or a structured query. One or more encoded values may indicate one or more selections from corresponding fixed set of possible attribute values (e.g., answers, choices, or options). The large language model operating as an encoder may be instructed to respond according to an encoding scheme, where encoded values may be assigned to different possible attribute values, or different possible responses to one or more

questions in the prompt about the context.

[0089] In the example above, the large language model is instructed to respond about three attributes/questions, where each attribute may have two possible attribute values, or each question may have two possible answers:

TABLE-US-00006 ATTRIBUTE POSSIBLE ATTRIBUTE VALUES whether user intent is to watch a free movie YES, NO whether user has mentioned an actor name in YES, NO the query whether user has mentioned a title in query YES, NO

[0090] Rather than responding using the attribute values, or in the form of a data structure, the large language model is instructed to respond using a binary string having one or more encoded values that have been assigned to the possible attribute values (e.g., according to an encoding scheme):

TABLE-US-00007 ENCODED VALUE(S) ASSIGNED TO ATTRIBUTE POSSIBLE ATTRIBUTE VALUES whether user intent is to watch a free movie “1” if YES “0” if NO whether user has mentioned an actor name “1” if YES “0” if NO in the query whether user has mentioned a title in query “1” if YES “0” if NO

[0091] The large language model generate a response having one or more encoded values in accordance with the encoding scheme. Possible responses (e.g., 8 possible responses) may include:

TABLE-US-00008 whether user intent whether user has whether user has Response having is to watch a free mentioned an actor mentioned a title in one or more movie name in the query query encoded values YES YES YES “111” YES YES NO “110” YES NO YES “101” YES NO NO “100” NO YES YES “011” NO YES NO “010” NO NO YES “001” NO NO NO “000”

[0092] In a different implementation, the large language model may have the following possible responses using ciphers or symbols in accordance with a different encoding scheme. In this encoding scheme, a cipher or symbol can indicate or encode a particular combination of answers. The cipher or symbol may indicate or encode a specific combination of attribute values or selections. Possible ciphers/symbols (e.g., 8 possible ciphers/symbols) may include:

TABLE-US-00009 Response having one whether user intent whether user has whether user has or more encoded is to watch a free mentioned an actor mentioned a title in values movie name in the query query (cipher/symbol) YES YES YES “A” YES YES NO “B” YES NO YES “C” YES NO NO “D” NO YES YES “E” NO YES NO “F” NO NO YES “G” NO NO NO “H”

[0093] In some embodiments, the instruction in the prompt to the large language model operating as an encoder may include a first question about the context with at least two or more possible answers. The one or more encoded values in the response generated by the large language model can indicate one of the at least two or more possible answers to the first question. For example, a response having “0” may indicate or encode an answer to the first question. In some embodiments, the instruction may further include a second question about the context with at least two or more further possible answers. The one or more encoded values in the response generated by the large language model can indicate one of the at least two or more possible answers to the first question and one of the at least two or more further possible answers to the second question. For example, a response having “01” may indicate or encode an answer to the first question and an answer to the second question.

[0094] In some embodiments, the one or more encoded value can include a first encoded value that indicate both the one of the at least two or more possible answers to the first question and the one of the at least two or more further possible answers. For example, a response having “B” may indicate or encode two or more answers to two or more questions in the prompt, provided that the encoding scheme encodes a combination of an answer to the first question and an answer to the second question using a single encoded value.

[0095] The encoding scheme may be applicable to other types of questions/attributes with more than two possible answers or possible attribute values. An encoding scheme can be implemented to encode the two or more possible answers by assigning appropriate encoding values to the two or

more possible answers.

[0096] In some embodiments, a response can include a binary string. The binary string may include one or more binary values, e.g., “010”.

[0097] In some embodiments, a response can include a numeric string. The numeric string may include one or more numbers, e.g., “412”.

[0098] In some embodiments, a response can include a character string. The character string may include one or more characters, e.g., “DA”.

[0099] In some embodiments, a response can include an alphanumeric string. The character string may include one or more characters, e.g., “1B”.

[0100] The one or more encoded values in the response may indicate one or more attributes about the context. The encoded values can be decoded (or translated) in accordance with the encoding scheme and used by one or more parts in a content item retrieval system. The encoded values may be used as control signals that may control one or more parts in a content item retrieval system.

[0101] In some embodiments, the one or more encoded values **410** may be used in a candidate generation part of a content item retrieval system (e.g., in candidate generation part **130** of the FIGS.). As illustrated, the one or more encoded values **410** may be provided to structured query builder **466** of a candidate generation part to generate a structured query. One or more encoded values **410** may correspond to one or more attributes about the context. The one or more attributes can include one or more intents **202**, e.g., an intent of the context usable by a retrieval engine, e.g., a retrieval engine in a candidate generation part, to filter content items. The one or more attributes can include a selection of one or more retrieval engine(s) **208**, e.g., an identification of the retrieval engine suitable for the context. The one or more attributes can include a selection of one or more content libraries **210**, e.g., an identification of the content library suitable for the context. The one or more attributes can include one or more regular expressions **206**.

[0102] Structured query builder **220** may receive, from the large language model, the response based on the prompt. The response may include one or more encoded values **410**. Structured query builder **220** may include encoded values to structured query translator **460**, which may decode the response having the one or more encoded values **410** (e.g., determine attributes corresponding to one or more encoded values **410**). Encoded values to structured query translator **460** may translate the one or more encoded values **410** into a structured query **292**. Encoded values to structured query translator **460** may apply the one or more encoded values **410** to structured query generation logic that produces structured queries conditioned on the one or more encoded values **410**. The structured query generation logic may include conditional logic or scripting that takes the one or more encoded values **410**, decode the one or more encoded values **410** according to the encoding scheme, and generates structured query **292**. The conditional logic can be very fast relative to the speed of a large language model generating the same structured query **292**. The structured query **292** may be input into a retrieval engine (e.g., a candidate generation part) to retrieve content items that correspond to the context from a content library. Structured query **292** may be used in candidate generation part as described in FIG. 2. The retrieved content items may be output to a user.

[0103] In some embodiments, the one or more encoded values may be used in a candidate ranking part of a content item retrieval system (e.g., in candidate ranking part **140** of the FIGS.). As illustrated, the one or more encoded values **420** may be provided to ranking parameter(s) processing part **488** of a candidate ranking part to augment ranking of relevant candidates. One or more encoded values **420** may correspond to one or more attributes about the context. The one or more attributes can include one or more intents **202**, e.g., an intent of the context usable by a ranking engine, e.g., a component in a candidate ranking part, to in scoring the content items. The one or more attributes can include one or more rows usable by the ranking engine in categorizing the content items.

[0104] Ranking parameter(s) processing part **488** may receive, from the large language model, the

response based on the prompt. The response may include one or more encoded values **420**.

Ranking parameter(s) processing part **488** may include encoded values to ranking parameter(s) translator **480**, which may decode the response having the one or more encoded values **420** (e.g., determine attributes corresponding to one or more encoded values **420**). Encoded values to ranking parameter(s) translator **480** may translate the one or more encoded values **420** into one or more ranking parameters **366**. Encoded values to ranking parameter(s) translator **480** may apply the one or more encoded values **410** to ranking parameters generation logic that produces ranking parameters conditioned on the one or more encoded values **420**. The ranking parameters generation logic may include conditional logic or scripting that takes the one or more encoded values **420**, decode the one or more encoded values **420** according to the encoding scheme, and generates one or more ranking parameters **366** or instructions to a ranking engine to use one or more ranking parameters **366**. The conditional logic can be very fast relative to a speed of a large language model generating the same one or more ranking parameters **366**. The one or more ranking parameters **366** may be input into a ranking engine (e.g., a component in a candidate ranking part) to rank content items that correspond to the context using the one or more ranking parameters **366**. One or more ranking parameters **366** may be used in candidate ranking part as described in FIG. 3. The ranked content items may be output to a user.

[0105] In some cases, candidate generation part **130** and/or candidate ranking part **140** may be implemented with logic that is controllable by one or more encoded values **410** and/or one or more encoded values **420**. Candidate generation part **130** and/or candidate ranking part **140** may receive one or more encoded values **410** and/or one or more encoded values **420** as control signals. Candidate generation part **130** may generate relevant candidates in a manner according to one or more encoded values **410**. Candidate ranking part **140** may rank relevant candidates according to one or more encoded values **420**.

Fine-Tuning a Large Language Model to Operate as an Encoder

[0106] FIG. 5 illustrates fine-tuning a large language model to output one or more encoded values, according to some embodiments of the disclosure. Off the shelf large language models are not trained or tuned to operate as an encoder. Off the shelf large language models may be trained or tuned to produce natural language responses, data structures, or structured queries. An off the shelf large language model may include previously trained large language model **504**, which may be trained using a large volume of training data **502**. Previously trained large language model **504** may generate predictions **506**. Predictions **506** may include one or more of: natural language responses, data structures, or structured queries.

[0107] Previously trained large language model **504** may undergo fine-tuning process **560**. Fine-tuning process **560** can fine-tune previously trained large language model **504** and turn previously trained large language model **504** into large language model as an encoder **534**. Encoder **534** may generate predictions **536**. Predictions **536** may include responses having one or more encoded values.

[0108] Fine-tuning process **560** may involve using training data **532**. Fine-tuning process **560** may include update **572** (e.g., an update function). Update **572** may update one or more (trained/learned) parameters of previously trained large language model **504** based on the predictions **536** made in response to training data **532**. Training data **532** may include prompts and ground-truth responses. Fine-tuning process **560** may further train previously trained large language model **504** (e.g., update one or more (trained/learned) parameters of previously trained large language model **504**) using a custom data set, e.g., training data **532**, to adapt previously trained large language model **504** to generate responses having one or more encoded values. Examples of fine-tuning process **560** may include parameter-efficient fine-tuning, low-rank adaptation of large language models, low-rank adaptation of large language models with quantized precision of weight parameters, updating of parameters with quantized precision, etc.

[0109] In some embodiments, prompts in training data **532** may be input into a previously trained

large language model **504**. A first prompt of the prompts can include a context having a natural language query for content items (e.g., a natural language query provided by a user, a natural language query generated by a generative machine learning model). The first prompt may include an instruction to produce a response about the context in an encoded output format. The instruction may prompt previously trained large language model **504** to follow an encoding scheme when generating a response to the first prompt. The context in the first prompt may include one or more contextual factors. Other prompts of the prompts may have similar properties of the first prompt, but with other natural language queries and optionally one or more other contextual factors. Update **572** may observe predictions **536** generated by the previously trained large language model (shown as large language model as an encoder **534** in training) in response to the prompts in training data **532**. Update **572** may compare the predictions **536** against ground-truth responses in training data **532** to the prompts. A first ground-truth response of the ground-truth responses comprises one or more encoded values corresponding to one or more attributes about the context. Other ground-truth responses of the ground-truth responses may have similar properties as the first ground-truth response. Update **572** may update one or more parameters of the previously trained large language model (shown as large language model as an encoder **534** in training) based on the comparing. [0110] Training data **532** for this specialized task of generating encoded values as responses may not be readily available. In some cases, procedures may be implemented to produce training data **532**.

[0111] In some cases, data having prompts and generated responses using previously trained large language model **504** may be processed so that the prompts include an instruction to generate responses in an encoded output format and that the responses are converted into an encoded output format. The prompts and the converted responses can be used as training data **532**.

[0112] One exemplary procedure to generate training data **532** includes convert generated data structures to encoded values **580**. Convert generated data structures to encoded values **580** may retrieve (past) structured data responses generated by the previously trained large language model **504** in response to further prompts. A first further prompt of the further prompts can include the context and a further instruction to produce a further response about the context in a structured data format. Convert generated data structures to encoded values **580** may convert the structured data responses into the ground-truth responses (e.g., by applying a suitable encoding scheme). Convert generated data structures to encoded values **580** may also modify the further prompts to include an instruction to produce a response about the context in an encoded output format. The modified prompts and the converted ground-truth responses in the encoded output format may be used as training data **532**.

[0113] In some cases, a (different) large language model can be used to generate natural language queries and optionally one or more contextual factors based on a set of attribute values. The generated natural language queries and optionally one or more generated contextual factors can be put into prompts that include an instruction to generate responses in an encoded output format. The set of attribute values may be converted into responses in an encoded output format. The prompts and the converted responses can be used as training data **532**.

[0114] Another exemplary procedure to generate training data **532** includes generate training data matching data structure **590**. Generate training data matching data structure **590** can leverage a large language model that may generate natural language responses to generate training data **532**. Generate training data matching data structure **590** can determine possible variations of structured data responses having different values for the one or more attributes. In some cases, generate training data matching data structure **590** possible variations of responses in an encoded output format, and apply a decoding function to produce the possible variations of structured data responses having different values for the one or more attributes. Generate training data matching data structure **590**, for a first possible variation of the possible variations having one or more first values to the one or more attributes, can input an instruction (or a prompt having the instruction) to

a further large language model to generate one or more contexts that exhibit the one or more first values to the one or more attributes. In other words, generate training data matching data structure **590** may instruct the further large language model to generate one or more natural language queries that may have or match the one or more attributes of a given possible data structure. The one or more generated contexts can include one or more natural language queries. The one or more generated contexts are to be used in the prompts in training data **532** to the previously trained large language model (shown as large language model as an encoder **534** in training). Generate training data matching data structure **590** may encode the one or more first values to the one or more attributes into a first ground-truth response of the ground-truth responses in training data **532**. Generate training data matching data structure **590** may convert the one or more first values to the one or more attributes into a response in an encoded output format. Multiple generated contexts and corresponding ground-truth response can be used as multiple entries in training data **532**.

Exemplary Methods for Content Item Retrieval

[0115] FIG. **6** is a flowchart showing method **600** for content item retrieval, according to some embodiments of the disclosure. Method **600** may be implemented in content item retrieval system **100**, e.g., in context understanding part **120** and candidate generation part **130**, in the FIGS.

[0116] In **602**, a context may be received. The context may include a natural language query from a user. An example of a context is illustrated in FIG. **1** as context **180**.

[0117] In **604**, a prompt may be input into a large language model, such as a large language model operating as an encoder as described herein. The prompt may include the context and an instruction to produce a response about the context in an encoded output format.

[0118] In **606**, the response generated based on the prompt may be received from the large language model. The response can include one or more encoded values corresponding to one or more attributes about the context.

[0119] In **608**, the one or more encoded values may be translated into a structured query.

[0120] In **610**, the structured query may be input into a retrieval engine to retrieve content items that correspond to the context from a content library.

[0121] In **612**, the content items can be output to the user.

[0122] FIG. **7** is a flowchart showing method **700** for ranking content items, according to some embodiments of the disclosure. Method **700** may be implemented in content item retrieval system **100**, e.g., in context understanding part **120** and candidate ranking part **140**, in the FIGS.

[0123] In **702**, a context may be received. The context may include a natural language query from a user. An example of a context is illustrated in FIG. **1** as context **180**.

[0124] In **704**, a prompt may be input into a large language model, such as a large language model operating as an encoder as described herein. The prompt may include the context and an instruction to produce a response about the context in an encoded output format.

[0125] In **706**, the response generated based on the prompt may be received from the large language model. The response can include one or more encoded values corresponding to one or more attributes about the context.

[0126] In **708**, the one or more encoded values may be translated into one or more ranking parameters.

[0127] In **710**, a ranking engine may rank content items that correspond to the context using the one or more ranking parameters.

[0128] In **712**, ranked content items may be output to the user.

Exemplary Methods for Fine-Tuning a Large Language Model

[0129] FIG. **8** is a flowchart showing method **800** for fine-tuning a large language model, according to some embodiments of the disclosure. Method **700** may be implemented using components illustrated in FIG. **5**.

[0130] In **802**, prompts may be input into a previously trained large language model. A first prompt of the prompts can include a context having a natural language query for content items, and an

instruction to produce a response about the context in an encoded output format.

[0131] In **804**, predictions generated by the previously trained large language model in response to the prompts can be observed.

[0132] In **806**, the predictions may be compared against ground-truth responses to the prompts. A first ground-truth response of the ground-truth responses can include one or more encoded values corresponding to one or more attributes about the context.

[0133] In **810**, one or more parameters of the previously trained large language model can be updated based on the comparing.

Exemplary Computing Device

[0134] FIG. **9** is a block diagram of an exemplary computing device **900**, according to some embodiments of the disclosure. One or more computing devices **900** may be used to implement the functionalities described with the FIGS. and herein. A number of components are illustrated in the FIGS. as included in the computing device **900**, but any one or more of these components may be omitted or duplicated, as suitable for the application. In some embodiments, some or all of the components included in the computing device **900** may be attached to one or more motherboards. In some embodiments, some or all of these components are fabricated onto a single system on a chip (SoC) die. Additionally, in various embodiments, the computing device **900** may not include one or more of the components illustrated in FIG. **8**, and the computing device **900** may include interface circuitry for coupling to the one or more components. For example, the computing device **900** may not include a display device **906**, and may include display device interface circuitry (e.g., a connector and driver circuitry) to which a display device **906** may be coupled. In another set of examples, the computing device **900** may not include an audio input device **918** or an audio output device **908** and may include audio input or output device interface circuitry (e.g., connectors and supporting circuitry) to which an audio input device **918** or audio output device **908** may be coupled.

[0135] The computing device **900** may include a processing device **902** (e.g., one or more processing devices, one or more of the same type of processing device, one or more of different types of processing device). The processing device **902** may include electronic circuitry that process electronic data from data storage elements (e.g., registers, memory, resistors, capacitors, quantum bit cells) to transform that electronic data into other electronic data that may be stored in registers and/or memory. Examples of processing device **902** may include a central processing unit (CPU), a graphics processing unit (GPU), a quantum processor, a machine learning processor, an artificial-intelligence processor, a neural network processor, an artificial-intelligence accelerator, an application specific integrated circuit (ASIC), an analog signal processor, an analog computer, a microprocessor, a digital signal processor, a field programmable gate array (FPGA), a tensor processing unit (TPU), a data processing unit (DPU), etc.

[0136] The computing device **900** may include a memory **904**, which may itself include one or more memory devices such as volatile memory (e.g., DRAM), nonvolatile memory (e.g., read-only memory (ROM)), high bandwidth memory (HBM), flash memory, solid state memory, and/or a hard drive. Memory **904** includes one or more non-transitory computer-readable storage media. In some embodiments, memory **904** may include memory that shares a die with the processing device **902**. In some embodiments, memory **904** includes one or more non-transitory computer-readable media storing instructions executable to perform operations described with the FIGS. and herein, such as the methods and operations illustrated in FIGS. **1-8**. Exemplary parts that may be encoded as instructions and stored in memory **904** are depicted. Parts may include one or more components in content item retrieval system **100**, such as context understanding part **120**, candidate generation part **130**, candidate ranking part **140**, structured query builder **466**, ranking parameter(s) processing part **488**, etc. Memory **904** may store instructions that encode one or more exemplary parts. The instructions stored in the one or more non-transitory computer-readable media may be executed by processing device **902**. In some embodiments, memory **904** may store data, e.g., data structures,

binary data, bits, metadata, files, blobs, etc., as described with the FIGS. and herein. Exemplary data that may be stored in memory **904** are depicted. Data may include one or more of: context **180**, results **106**, one or more encoded values **410** and one or more encoded values **420**. Data may include one or more of: structured query **292** and one or more ranking parameters **366**. Memory **904** may store one or more data as depicted.

[0137] In some embodiments, memory **904** may store one or more machine learning models (and or parts thereof), such as a large language model as described herein (e.g., previously trained large language model **504**, large language model as an encoder **534**, and large language model in generate training data matching data structure **590**). Memory **904** may store training data such as training data **502** and training data **532**. Memory **904** may store instructions that perform operations associated with fine-tuning process **560** and update **572** of FIG. 5. Memory **904** may store input data, output data, intermediate outputs, intermediate inputs of one or more machine learning models. Memory **904** may store instructions to perform one or more operations of the machine learning model. Memory **904** may store one or more parameters used by the machine learning model. Memory **904** may store information that encodes how processing units of the machine learning model are connected with each other. Memory **904** may store instructions that perform operations associated with convert generated data structures to encoded values **580**. Memory **904** may store instructions that perform operations associated with generate training data matching data structure **590**.

[0138] In some embodiments, the computing device **900** may include a communication device **912** (e.g., one or more communication devices). For example, the communication device **912** may be configured for managing wired and/or wireless communications for the transfer of data to and from the computing device **900**. The term “wireless” and its derivatives may be used to describe circuits, devices, systems, methods, techniques, communications channels, etc., that may communicate data through the use of modulated electromagnetic radiation through a nonsolid medium. The term does not imply that the associated devices do not contain any wires, although in some embodiments they might not. The communication device **912** may implement any of a number of wireless standards or protocols, including but not limited to Institute for Electrical and Electronic Engineers (IEEE) standards including Wi-Fi (IEEE 802.10 family), IEEE 802.16 standards (e.g., IEEE 802.16-2005 Amendment), Long-Term Evolution (LTE) project along with any amendments, updates, and/or revisions (e.g., advanced LTE project, ultramobile broadband (UMB) project (also referred to as “3GPP2”), etc.). IEEE 802.16 compatible Broadband Wireless Access (BWA) networks are generally referred to as WiMAX networks, an acronym that stands for worldwide interoperability for microwave access, which is a certification mark for products that pass conformity and interoperability tests for the IEEE 802.16 standards. The communication device **912** may operate in accordance with a Global System for Mobile Communication (GSM), General Packet Radio Service (GPRS), Universal Mobile Telecommunications System (UMTS), High Speed Packet Access (HSPA), Evolved HSPA (E-HSPA), or LTE network. The communication device **912** may operate in accordance with Enhanced Data for GSM Evolution (EDGE), GSM EDGE Radio Access Network (GERAN), Universal Terrestrial Radio Access Network (UTRAN), or Evolved UTRAN (E-UTRAN). The communication device **912** may operate in accordance with Code-division Multiple Access (CDMA), Time Division Multiple Access (TDMA), Digital Enhanced Cordless Telecommunications (DECT), Evolution-Data Optimized (EV-DO), and derivatives thereof, as well as any other wireless protocols that are designated as 3G, 4G, 5G, and beyond. The communication device **912** may operate in accordance with other wireless protocols in other embodiments. The computing device **900** may include an antenna **922** to facilitate wireless communications and/or to receive other wireless communications (such as radio frequency transmissions). The computing device **900** may include receiver circuits and/or transmitter circuits. In some embodiments, the communication device **912** may manage wired communications, such as electrical, optical, or any other suitable communication protocols (e.g., the Ethernet). As noted above, the communication

device **912** may include multiple communication chips. For instance, a first communication device **912** may be dedicated to shorter-range wireless communications such as Wi-Fi or Bluetooth, and a second communication device **912** may be dedicated to longer-range wireless communications such as global positioning system (GPS), EDGE, GPRS, CDMA, WiMAX, LTE, EV-DO, or others. In some embodiments, a first communication device **912** may be dedicated to wireless communications, and a second communication device **912** may be dedicated to wired communications.

[0139] The computing device **900** may include power source/power circuitry **914**. The power source/power circuitry **914** may include one or more energy storage devices (e.g., batteries or capacitors) and/or circuitry for coupling components of the computing device **900** to an energy source separate from the computing device **900** (e.g., DC power, AC power, etc.).

[0140] The computing device **900** may include a display device **906** (or corresponding interface circuitry, as discussed above). The display device **906** may include any visual indicators, such as a heads-up display, a computer monitor, a projector, a touchscreen display, a liquid crystal display (LCD), a light-emitting diode display, or a flat panel display, for example.

[0141] The computing device **900** may include an audio output device **908** (or corresponding interface circuitry, as discussed above). The audio output device **908** may include any device that generates an audible indicator, such as speakers, headsets, or earbuds, for example.

[0142] The computing device **900** may include an audio input device **918** (or corresponding interface circuitry, as discussed above). The audio input device **918** may include any device that generates a signal representative of a sound, such as microphones, microphone arrays, or digital instruments (e.g., instruments having a musical instrument digital interface (MIDI) output).

[0143] The computing device **900** may include a GPS device **916** (or corresponding interface circuitry, as discussed above). The GPS device **916** may be in communication with a satellite-based system and may receive a location of the computing device **900**, as known in the art.

[0144] The computing device **900** may include a sensor **930** (or one or more sensors). The computing device **900** may include corresponding interface circuitry, as discussed above). Sensor **930** may sense physical phenomenon and translate the physical phenomenon into electrical signals that can be processed by, e.g., processing device **902**. Examples of sensor **930** may include: capacitive sensor, inductive sensor, resistive sensor, electromagnetic field sensor, light sensor, camera, imager, microphone, pressure sensor, temperature sensor, vibrational sensor, accelerometer, gyroscope, strain sensor, moisture sensor, humidity sensor, distance sensor, range sensor, time-of-flight sensor, pH sensor, particle sensor, air quality sensor, chemical sensor, gas sensor, biosensor, ultrasound sensor, a scanner, etc.

[0145] The computing device **900** may include another output device **910** (or corresponding interface circuitry, as discussed above). Examples of the other output device **910** may include an audio codec, a video codec, a printer, a wired or wireless transmitter for providing information to other devices, haptic output device, gas output device, vibrational output device, lighting output device, home automation controller, or an additional storage device.

[0146] The computing device **900** may include another input device **920** (or corresponding interface circuitry, as discussed above). Examples of the other input device **920** may include an accelerometer, a gyroscope, a compass, an image capture device, a keyboard, a cursor control device such as a mouse, a stylus, a touchpad, a bar code reader, a Quick Response (QR) code reader, any sensor, or a radio frequency identification (RFID) reader.

[0147] The computing device **900** may have any desired form factor, such as a handheld or mobile computer system (e.g., a cell phone, a smart phone, a mobile internet device, a music player, a tablet computer, a laptop computer, a netbook computer, a personal digital assistant (PDA), an ultramobile personal computer, a remote control, wearable device, headgear, eyewear, footwear, electronic clothing, etc.), a desktop computer system, a server or other networked computing component, a printer, a scanner, a monitor, a set-top box, an entertainment control unit, a vehicle

control unit, a digital camera, a digital video recorder, an Internet-of-Things device (e.g., light bulb, cable, power plug, power source, lighting system, audio assistant, audio speaker, smart home device, smart thermostat, camera monitor device, sensor device, smart home doorbell, motion sensor device), a virtual reality system, an augmented reality system, a mixed reality system, or a wearable computer system. In some embodiments, the computing device **900** may be any other electronic device that processes data.

Select Examples

[0148] Example 1 provides a method, including receiving a context, where the context includes a natural language query from a user; inputting, into a large language model, a prompt including the context and an instruction to produce a response about the context in an encoded output format; receiving, from the large language model, the response generated based on the prompt, where the response includes one or more encoded values corresponding to one or more attributes about the context; translating the one or more encoded values into a structured query; inputting the structured query into a retrieval engine to retrieve content items that correspond to the context from a content library; and outputting the content items to the user.

[0149] Example 2 provides the method of example 1, where the context includes one or more contextual factors about the user.

[0150] Example 3 provides the method of example 1 or 2, where the context includes one or more contextual factors about a device used by the user.

[0151] Example 4 provides the method of any one of examples 1-3, where: the instruction includes a first question about the context with at least two or more possible answers; and the one or more encoded values in the response indicate one of the at least two or more possible answers to the first question.

[0152] Example 5 provides the method of example 4, where: the instruction includes a second question about the context with at least two or more further possible answers; and the one or more encoded values in the response indicate one of the at least two or more possible answers to the first question and one of the at least two or more further possible answers.

[0153] Example 6 provides the method of example 5, where the one or more encoded values include a first encoded value that indicates both the one of the at least two or more possible answers to the first question and the one of the at least two or more further possible answers to the second question.

[0154] Example 7 provides the method of any one of examples 1-6, where the encoded output format is a binary string, and the one or more encoded values include one or more binary values.

[0155] Example 8 provides the method of any one of examples 1-7, where the one or more attributes includes an intent of the context usable by the retrieval engine to filter content items.

[0156] Example 9 provides the method of any one of examples 1-8, where the one or more attributes includes an identification of the retrieval engine suitable for the context.

[0157] Example 10 provides the method of any one of examples 1-9, where the one or more attributes includes an identification of the content library suitable for the context.

[0158] Example 11 provides the method of any one of examples 1-10, where translating the one or more encoded values into the structured query includes applying the one or more encoded values to structured query generation logic that produces structured queries conditioned on the one or more encoded values.

[0159] Example 12 provides the method of any one of examples 1-11, where the response does not include natural language text.

[0160] Example 13 provides the method of any one of examples 1-12, where the response does not include one or more attribute-value pairs.

[0161] Example 14 provides a method, including receiving a context, where the context includes a natural language query from a user; inputting, into a large language model, a prompt including the context and an instruction to produce a response about the context in an encoded output format;

receiving, from the large language model, the response generated based on the prompt, where the response includes one or more encoded values corresponding to one or more attributes about the context; translating the one or more encoded values into one or more ranking parameters; ranking, by a ranking engine, content items that correspond to the context using the one or more ranking parameters; and outputting ranked content items to the user.

[0162] Example 15 provides the method of example 14, where the context includes one or more contextual factors about the user.

[0163] Example 16 provides the method of example 14 or 15, where the context includes one or more contextual factors about a device used by the user.

[0164] Example 17 provides the method of any one of examples 14-16, where: the instruction includes a first question about the context with at least two or more possible answers; and the one or more encoded values in the response indicate one of the at least two or more possible answers to the first question.

[0165] Example 18 provides the method of example 17, where: the instruction includes a second question about the context with at least two or more further possible answers; and the one or more encoded values in the response indicate one of the at least two or more possible answers to the first question and one of the at least two or more further possible answers to the second question.

[0166] Example 19 provides the method of example 18, where the one or more encoded values include a first encoded value that indicate both the one of the at least two or more possible answers to the first question and the one of the at least two or more further possible answers.

[0167] Example 20 provides the method of any one of examples 14-19, where the encoded output format is a binary string, and the one or more encoded values include one or more binary values.

[0168] Example 21 provides the method of any one of examples 14-20, where the one or more attributes includes an intent of the context usable by the ranking engine in scoring the content items.

[0169] Example 22 provides the method of any one of examples 14-21, where the one or more attributes includes one or more rows usable by the ranking engine in categorizing the content items.

[0170] Example 23 provides the method of any one of examples 14-22, where translating the one or more encoded values into the one or more ranking parameters includes applying the one or more encoded values to ranking parameters generation logic that produces ranking parameters conditioned on the one or more encoded values.

[0171] Example 24 provides the method of any one of examples 14-23, where the response does not include natural language text.

[0172] Example 25 provides the method of any one of examples 14-24, where the response does not include one or more attribute-value pairs.

[0173] Example 26 provides a method, including inputting prompts into a previously trained large language model, where a first prompt of the prompts includes a context having a natural language query for content items, and an instruction to produce a response about the context in an encoded output format; observing predictions generated by the previously trained large language model in response to the prompts; comparing the predictions against ground-truth responses to the prompts, where a first ground-truth response of the ground-truth responses includes one or more encoded values corresponding to one or more attributes about the context; and updating one or more parameters of the previously trained large language model based on the comparing.

[0174] Example 27 provides the method of example 26, where the context further includes one or more contextual factors.

[0175] Example 28 provides the method of example 26 or 27, further including retrieving structured data responses generated by the previously trained large language model in response to further prompts, where a first further prompt of the further prompts has the context and a further instruction to produce a further response about the context in a structured data format; and converting the structured data responses into the ground-truth responses.

[0176] Example 29 provides the method of any one of examples 26-28, further including determining possible variations of structured data responses having different values for the one or more attributes; and for a first possible variation of the possible variations having one or more first values to the one or more attributes, inputting an instruction to a further large language model to generate one or more contexts that exhibit the one or more first values to the one or more attributes, where the one or more generated contexts include one or more natural language queries, and the one or more generated contexts are to be used in the prompts to the previously trained large language model; and encoding the one or more first values to the one or more attributes into a first ground-truth response of the ground-truth responses.

[0177] Example 30 provides one or more non-transitory computer-readable media storing instructions that, when executed by one or more processors, cause the one or more processors to: receive a context, where the context includes a natural language query from a user; input, into a large language model, a prompt including the context and an instruction to produce a response about the context in an encoded output format; receive, from the large language model, the response generated based on the prompt, where the response includes one or more encoded values corresponding to one or more attributes about the context; translate the one or more encoded values into a structured query; input the structured query into a retrieval engine to retrieve content items that correspond to the context from a content library; and output the content items to the user.

[0178] Example 31 provides the one or more non-transitory computer-readable media of example 30, where the context includes one or more contextual factors about the user.

[0179] Example 32 provides the one or more non-transitory computer-readable media of example 30 or 31, where the context includes one or more contextual factors about a device used by the user.

[0180] Example 33 provides the one or more non-transitory computer-readable media of any one of examples 30-32, where: the instruction includes a first question about the context with at least two or more possible answers; and the one or more encoded values in the response indicate one of the at least two or more possible answers to the first question.

[0181] Example 34 provides the one or more non-transitory computer-readable media of example 33, where: the instruction includes a second question about the context with at least two or more further possible answers; and the one or more encoded values in the response indicate one of the at least two or more possible answers to the first question and one of the at least two or more further possible answers.

[0182] Example 35 provides the one or more non-transitory computer-readable media of example 34, where the one or more encoded values include a first encoded value that indicates both the one of the at least two or more possible answers to the first question and the one of the at least two or more further possible answers to the second question.

[0183] Example 36 provides the one or more non-transitory computer-readable media of any one of examples 30-35, where the encoded output format is a binary string, and the one or more encoded values include one or more binary values.

[0184] Example 37 provides the one or more non-transitory computer-readable media of any one of examples 30-36, where the one or more attributes includes an intent of the context usable by the retrieval engine to filter content items.

[0185] Example 38 provides the one or more non-transitory computer-readable media of any one of examples 30-37, where the one or more attributes includes an identification of the retrieval engine suitable for the context.

[0186] Example 39 provides the one or more non-transitory computer-readable media of any one of examples 30-38, where the one or more attributes includes an identification of the content library suitable for the context.

[0187] Example 40 provides the one or more non-transitory computer-readable media of any one of examples 30-39, where translating the one or more encoded values into the structured query includes applying the one or more encoded values to structured query generation logic that

produces structured queries conditioned on the one or more encoded values.

[0188] Example 41 provides the one or more non-transitory computer-readable media of any one of examples 30-40, where the response does not include natural language text.

[0189] Example 42 provides the one or more non-transitory computer-readable media of any one of examples 30-41, where the response does not include one or more attribute-value pairs.

[0190] Example 43 provides one or more non-transitory computer-readable media storing instructions that, when executed by one or more processors, cause the one or more processors to: receive a context, where the context includes a natural language query from a user; input, into a large language model, a prompt including the context and an instruction to produce a response about the context in an encoded output format; receive, from the large language model, the response generated based on the prompt, where the response includes one or more encoded values corresponding to one or more attributes about the context; translate the one or more encoded values into one or more ranking parameters; rank, by a ranking engine, content items that correspond to the context using the one or more ranking parameters; and output ranked content items to the user.

[0191] Example 44 provides the one or more non-transitory computer-readable media of example 43, where the context includes one or more contextual factors about the user.

[0192] Example 45 provides the one or more non-transitory computer-readable media of example 43 or 44, where the context includes one or more contextual factors about a device used by the user.

[0193] Example 46 provides the one or more non-transitory computer-readable media of any one of examples 43-45, where: the instruction includes a first question about the context with at least two or more possible answers; and the one or more encoded values in the response indicate one of the at least two or more possible answers to the first question.

[0194] Example 47 provides the one or more non-transitory computer-readable media of example 46, where: the instruction includes a second question about the context with at least two or more further possible answers; and the one or more encoded values in the response indicate one of the at least two or more possible answers to the first question and one of the at least two or more further possible answers to the second question.

[0195] Example 48 provides the one or more non-transitory computer-readable media of example 47, where the one or more encoded values include a first encoded value that indicate both the one of the at least two or more possible answers to the first question and the one of the at least two or more further possible answers.

[0196] Example 49 provides the one or more non-transitory computer-readable media of any one of examples 43-48, where the encoded output format is a binary string, and the one or more encoded values include one or more binary values.

[0197] Example 50 provides the one or more non-transitory computer-readable media of any one of examples 43-49, where the one or more attributes includes an intent of the context usable by the ranking engine in scoring the content items.

[0198] Example 51 provides the one or more non-transitory computer-readable media of any one of examples 43-50, where the one or more attributes includes one or more rows usable by the ranking engine in categorizing the content items.

[0199] Example 52 provides the one or more non-transitory computer-readable media of any one of examples 43-51, where translating the one or more encoded values into the one or more ranking parameters includes applying the one or more encoded values to ranking parameters generation logic that produces ranking parameters conditioned on the one or more encoded values.

[0200] Example 53 provides the one or more non-transitory computer-readable media of any one of examples 43-52, where the response does not include natural language text.

[0201] Example 54 provides the one or more non-transitory computer-readable media of any one of examples 43-53, where the response does not include one or more attribute-value pairs.

[0202] Example 55 provides one or more non-transitory computer-readable media storing instructions that, when executed by one or more processors, cause the one or more processors to:

input prompts into a previously trained large language model, where a first prompt of the prompts includes a context having a natural language query for content items, and an instruction to produce a response about the context in an encoded output format; observe predictions generated by the previously trained large language model in response to the prompts; compare the predictions against ground-truth responses to the prompts, where a first ground-truth response of the ground-truth responses includes one or more encoded values corresponding to one or more attributes about the context; and update one or more parameters of the previously trained large language model based on the comparing.

[0203] Example 56 provides the one or more non-transitory computer-readable media of example 55, where the context further includes one or more contextual factors.

[0204] Example 57 provides the one or more non-transitory computer-readable media of example 55 or 56, where the instructions further cause the one or more processors to: retrieve structured data responses generated by the previously trained large language model in response to further prompts, where a first further prompt of the further prompts has the context and a further instruction to produce a further response about the context in a structured data format; and convert the structured data responses into the ground-truth responses.

[0205] Example 58 provides the one or more non-transitory computer-readable media of any one of examples 55-57, where the instructions further cause the one or more processors to: determine possible variations of structured data responses having different values for the one or more attributes; and for a first possible variation of the possible variations having one or more first values to the one or more attributes, input an instruction to a further large language model to generate one or more contexts that exhibit the one or more first values to the one or more attributes, where the one or more generated contexts include one or more natural language queries, and the one or more generated contexts are to be used in the prompts to the previously trained large language model; and encode the one or more first values to the one or more attributes into a first ground-truth response of the ground-truth responses.

[0206] Example 59 provides a system including one or more processors; and one or more non-transitory computer-readable media storing instructions that, when executed by the one or more processors, cause the one or more processors to: receive a context, where the context includes a natural language query from a user; input, into a large language model, a prompt including the context and an instruction to produce a response about the context in an encoded output format; receive, from the large language model, the response generated based on the prompt, where the response includes one or more encoded values corresponding to one or more attributes about the context; translate the one or more encoded values into a structured query; input the structured query into a retrieval engine to retrieve content items that correspond to the context from a content library; and output the content items to the user.

[0207] Example 60 provides the system of example 59, where the context includes one or more contextual factors about the user.

[0208] Example 61 provides the system of example 59 or 60, where the context includes one or more contextual factors about a device used by the user.

[0209] Example 62 provides the system of any one of examples 59-61, where: the instruction includes a first question about the context with at least two or more possible answers; and the one or more encoded values in the response indicate one of the at least two or more possible answers to the first question.

[0210] Example 63 provides the system of example 62, where: the instruction includes a second question about the context with at least two or more further possible answers; and the one or more encoded values in the response indicate one of the at least two or more possible answers to the first question and one of the at least two or more further possible answers.

[0211] Example 64 provides the system of example 63, where the one or more encoded values include a first encoded value that indicates both the one of the at least two or more possible

answers to the first question and the one of the at least two or more further possible answers to the second question.

[0212] Example 65 provides the system of any one of examples 59-64, where the encoded output format is a binary string, and the one or more encoded values include one or more binary values.

[0213] Example 66 provides the system of any one of examples 59-65, where the one or more attributes includes an intent of the context usable by the retrieval engine to filter content items.

[0214] Example 67 provides the system of any one of examples 59-66, where the one or more attributes includes an identification of the retrieval engine suitable for the context.

[0215] Example 68 provides the system of any one of examples 59-67, where the one or more attributes includes an identification of the content library suitable for the context.

[0216] Example 69 provides the system of any one of examples 59-68, where translating the one or more encoded values into the structured query includes applying the one or more encoded values to structured query generation logic that produces structured queries conditioned on the one or more encoded values.

[0217] Example 70 provides the system of any one of examples 59-69, where the response does not include natural language text.

[0218] Example 71 provides the system of any one of examples 59-70, where the response does not include one or more attribute-value pairs.

[0219] Example 72 provides a system including one or more processors; and one or more non-transitory computer-readable media storing instructions that, when executed by the one or more processors, cause the one or more processors to: receive a context, where the context includes a natural language query from a user; input, into a large language model, a prompt including the context and an instruction to produce a response about the context in an encoded output format; receive, from the large language model, the response generated based on the prompt, where the response includes one or more encoded values corresponding to one or more attributes about the context; translate the one or more encoded values into one or more ranking parameters; rank, by a ranking engine, content items that correspond to the context using the one or more ranking parameters; and output ranked content items to the user.

[0220] Example 73 provides the system of example 72, where the context includes one or more contextual factors about the user.

[0221] Example 74 provides the system of example 72 or 73, where the context includes one or more contextual factors about a device used by the user.

[0222] Example 75 provides the system of any one of examples 72-74, where: the instruction includes a first question about the context with at least two or more possible answers; and the one or more encoded values in the response indicate one of the at least two or more possible answers to the first question.

[0223] Example 76 provides the system of example 75, where: the instruction includes a second question about the context with at least two or more further possible answers; and the one or more encoded values in the response indicate one of the at least two or more possible answers to the first question and one of the at least two or more further possible answers to the second question.

[0224] Example 77 provides the system of example 76, where the one or more encoded values include a first encoded value that indicate both the one of the at least two or more possible answers to the first question and the one of the at least two or more further possible answers.

[0225] Example 78 provides the system of any one of examples 72-77, where the encoded output format is a binary string, and the one or more encoded values include one or more binary values.

[0226] Example 79 provides the system of any one of examples 72-78, where the one or more attributes includes an intent of the context usable by the ranking engine in scoring the content items.

[0227] Example 80 provides the system of any one of examples 72-79, where the one or more attributes includes one or more rows usable by the ranking engine in categorizing the content items.

[0228] Example 81 provides the system of any one of examples 72-80, where translating the one or more encoded values into the one or more ranking parameters includes applying the one or more encoded values to ranking parameters generation logic that produces ranking parameters conditioned on the one or more encoded values.

[0229] Example 82 provides the system of any one of examples 72-81, where the response does not include natural language text.

[0230] Example 83 provides the system of any one of examples 72-82, where the response does not include one or more attribute-value pairs.

[0231] Example 84 provides a system including one or more processors; and one or more non-transitory computer-readable media storing instructions that, when executed by the one or more processors, cause the one or more processors to: input prompts into a previously trained large language model, where a first prompt of the prompts includes a context having a natural language query for content items, and an instruction to produce a response about the context in an encoded output format; observe predictions generated by the previously trained large language model in response to the prompts; compare the predictions against ground-truth responses to the prompts, where a first ground-truth response of the ground-truth responses includes one or more encoded values corresponding to one or more attributes about the context; and update one or more parameters of the previously trained large language model based on the comparing.

[0232] Example 85 provides the system of example 84, where the context further includes one or more contextual factors.

[0233] Example 86 provides the system of example 84 or 85, where the instructions further cause the one or more processors to: retrieve structured data responses generated by the previously trained large language model in response to further prompts, where a first further prompt of the further prompts has the context and a further instruction to produce a further response about the context in a structured data format; and convert the structured data responses into the ground-truth responses.

[0234] Example 87 provides the system of any one of examples 84-86, where the instructions further cause the one or more processors to: determine possible variations of structured data responses having different values for the one or more attributes; and for a first possible variation of the possible variations having one or more first values to the one or more attributes, input an instruction to a further large language model to generate one or more contexts that exhibit the one or more first values to the one or more attributes, where the one or more generated contexts include one or more natural language queries, and the one or more generated contexts are to be used in the prompts to the previously trained large language model; and encode the one or more first values to the one or more attributes into a first ground-truth response of the ground-truth responses.

[0235] Example A provides one or more non-transitory computer-readable media storing instructions that, when executed by one or more processors, cause the one or more processors to perform any one of the methods provided in examples 1-29 and FIGS. **6-8**.

[0236] Example B provides an apparatus comprising means to carry out or means for carrying out any one of the computer-implemented methods provided in examples 1-29 and FIGS. **6-8**.

[0237] Example C provides a computer-implemented system, comprising one or more processors, and one or more non-transitory computer-readable media storing instructions that, when executed by the one or more processors, cause the one or more processors to perform any one of the methods provided in examples 1-29 and FIGS. **6-8**.

[0238] Example D provides a computer-implemented system comprising one or more components illustrated in any one of FIGS. **1-5** and **9** to perform operations described herein.

Variations and Other Notes

[0239] Although the operations of the example methods shown in and described with reference to the FIGS. are illustrated as occurring once each and in a particular order, it will be recognized that the operations may be performed in any suitable order and repeated as desired. Additionally, one or

more operations may be performed in parallel. Furthermore, the operations illustrated in the FIGS. may be combined or may include more or fewer details than described.

[0240] A large language model, as used herein, describes a machine learning model, or a deep learning model. A large language model may include many neural network layers, such as recurrent layers, feedforward layers, embedding layers, and attention layers. A neural network layer may include one or more neurons or nodes, where a neuron or node may perform a function and/or operation on input(s) to generate output(s) according to one or more (trained/learned) parameters. These neural network layers work together to process the input text and generate output content by making predictions of the next token in the output content. The embedding layer can create embeddings from the input text, while the attention layer may weigh the importance of different tokens in a sequence, allowing the large language model to capture dependencies and relationships. A recurrent layer is a type of neural network layer that allows the model to process sequences of variable length. It does this by maintaining a hidden state that is updated at each time step, allowing the model to remember information from previous time steps. The feedforward layer is a type of neural network layer that applies a linear transformation to the input data, followed by a non-linear activation function. The embedding layer is a type of neural network layer that maps discrete tokens to continuous vectors, allowing the model to process text data. The attention layer is a type of neural network layer that allows the model to focus on different parts of the input sequence, depending on the task at hand. The attention layer may be transformer-based. The transformer architecture introduced parallel processing to sequences and effectively addressed the issues of long-range dependencies through self-attention mechanisms. The attention mechanism computes a weighted sum of input values (or values from the previous layer), where the weights are decided based on the query, key, and value representations of the data. In the context of large language models, the attention layer is a component that enables the model to capture long-range dependencies and contextual information. The attention mechanism enables the model to focus on specific parts of the input text that are relevant to the task at hand, allowing it to generate more accurate outputs.

[0241] The above description of illustrated implementations of the disclosure, including what is described in the Abstract, is not intended to be exhaustive or to limit the disclosure to the precise forms disclosed. While specific implementations of, and examples for, the disclosure are described herein for illustrative purposes, various equivalent modifications are possible within the scope of the disclosure, as those skilled in the relevant art will recognize. These modifications may be made to the disclosure in light of the above detailed description.

[0242] For purposes of explanation, specific numbers, materials and configurations are set forth in order to provide a thorough understanding of the illustrative implementations. However, it will be apparent to one skilled in the art that the present disclosure may be practiced without the specific details and/or that the present disclosure may be practiced with only some of the described aspects. In other instances, well known features are omitted or simplified in order not to obscure the illustrative implementations.

[0243] Further, references are made to the accompanying drawings that form a part hereof, and in which are shown, by way of illustration, embodiments that may be practiced. It is to be understood that other embodiments may be utilized, and structural or logical changes may be made without departing from the scope of the present disclosure. Therefore, the following detailed description is not to be taken in a limiting sense.

[0244] Various operations may be described as multiple discrete actions or operations in turn, in a manner that is most helpful in understanding the disclosed subject matter. However, the order of description should not be construed as to imply that these operations are necessarily order dependent. In particular, these operations may not be performed in the order of presentation. Operations described may be performed in a different order from the described embodiment. Various additional operations may be performed or described operations may be omitted in

additional embodiments.

[0245] For the purposes of the present disclosure, the phrase “A or B” or the phrase “A and/or B” means (A), (B), or (A and B). For the purposes of the present disclosure, the phrase “A, B, or C” or the phrase “A, B, and/or C” means (A), (B), (C), (A and B), (A and C), (B and C), or (A, B, and C). The term “between,” when used with reference to measurement ranges, is inclusive of the ends of the measurement ranges.

[0246] The description uses the phrases “in an embodiment” or “in embodiments,” which may each refer to one or more of the same or different embodiments. The terms “comprising,” “including,” “having,” and the like, as used with respect to embodiments of the present disclosure, are synonymous. The disclosure may use perspective-based descriptions such as “above,” “below,” “top,” “bottom,” and “side” to explain various features of the drawings, but these terms are simply for ease of discussion, and do not imply a desired or required orientation. The accompanying drawings are not necessarily drawn to scale. Unless otherwise specified, the use of the ordinal adjectives “first,” “second,” and “third,” etc., to describe a common object, merely indicates that different instances of like objects are being referred to and are not intended to imply that the objects so described must be in a given sequence, either temporally, spatially, in ranking or in any other manner.

[0247] In the following detailed description, various aspects of the illustrative implementations will be described using terms commonly employed by those skilled in the art to convey the substance of their work to others skilled in the art.

[0248] The terms “substantially,” “close,” “approximately,” “near,” and “about,” generally refer to being within $\pm 20\%$ of a target value as described herein or as known in the art. Similarly, terms indicating orientation of various elements, e.g., “coplanar,” “perpendicular,” “orthogonal,” “parallel,” or any other angle between the elements, generally refer to being within $\pm 5\text{--}20\%$ of a target value as described herein or as known in the art.

[0249] In addition, the terms “comprise,” “comprising,” “include,” “including,” “have,” “having” or any other variation thereof, are intended to cover a non-exclusive inclusion. For example, a method, process, or device, that comprises a list of elements is not necessarily limited to only those elements but may include other elements not expressly listed or inherent to such method, process, or device. Also, the term “or” refers to an inclusive “or” and not to an exclusive “or.”

[0250] The systems, methods and devices of this disclosure each have several innovative aspects, no single one of which is solely responsible for all desirable attributes disclosed herein. Details of one or more implementations of the subject matter described in this specification are set forth in the description and the accompanying drawings.

Claims

1. A method, comprising: receiving a context, wherein the context includes a natural language query from a user; inputting, into a large language model, a prompt comprising the context and an instruction to produce a response about the context in an encoded output format; receiving, from the large language model, the response generated based on the prompt, wherein the response comprises one or more encoded values corresponding to one or more attributes about the context; translating the one or more encoded values into a structured query; inputting the structured query into a retrieval engine to retrieve content items that correspond to the context from a content library; and outputting the content items to the user.
2. The method of claim 1, wherein the context includes one or more contextual factors about the user.
3. The method of claim 1, wherein the context includes one or more contextual factors about a device used by the user.
4. The method of claim 1, wherein: the instruction comprises a first question about the context with

- at least two or more possible answers; and the one or more encoded values in the response indicate one of the at least two or more possible answers to the first question.
- 5.** The method of claim 4, wherein: the instruction comprises a second question about the context with at least two or more further possible answers; and the one or more encoded values in the response indicate one of the at least two or more possible answers to the first question and one of the at least two or more further possible answers.
- 6.** The method of claim 5, wherein the one or more encoded values comprise a first encoded value that indicates both the one of the at least two or more possible answers to the first question and the one of the at least two or more further possible answers to the second question.
- 7.** The method of claim 1, wherein the encoded output format is a binary string, and the one or more encoded values comprise one or more binary values.
- 8.** The method of claim 1, wherein the one or more attributes comprises: an intent of the context usable by the retrieval engine to filter content items.
- 9.** The method of claim 1, wherein the one or more attributes comprises: an identification of the retrieval engine suitable for the context.
- 10.** The method of claim 1, wherein the one or more attributes comprises: an identification of the content library suitable for the context.
- 11.** The method of claim 1, wherein translating the one or more encoded values into the structured query comprises: applying the one or more encoded values to structured query generation logic that produces structured queries conditioned on the one or more encoded values.
- 12.** The method of claim 1, wherein the response does not include natural language text.
- 13.** The method of claim 1, wherein the response does not include one or more attribute-value pairs.
- 14.** One or more non-transitory computer-readable media storing instructions that, when executed by one or more processors, cause the one or more processors to: receive a context, wherein the context includes a natural language query from a user; input, into a large language model, a prompt comprising the context and an instruction to produce a response about the context in an encoded output format, wherein the encoded output format is a binary string; receive, from the large language model, the response generated based on the prompt, wherein the response comprises one or more encoded values corresponding to one or more attributes about the context, and the one or more encoded values comprise one or more binary values; translate the one or more encoded values into a structured query; input the structured query into a retrieval engine to retrieve content items that correspond to the context from a content library; and output the content items to the user.
- 15.** The one or more non-transitory computer-readable media of claim 14, wherein: the instruction comprises a first question about the context with at least two or more possible answers; and the one or more encoded values in the response indicate one of the at least two or more possible answers to the first question.
- 16.** The one or more non-transitory computer-readable media of claim 15, wherein: the instruction comprises a second question about the context with at least two or more further possible answers; and the one or more encoded values in the response indicate one of the at least two or more possible answers to the first question and one of the at least two or more further possible answers.
- 17.** The one or more non-transitory computer-readable media of claim 16, wherein the one or more encoded values comprise a first encoded value that indicates both the one of the at least two or more possible answers to the first question and the one of the at least two or more further possible answers to the second question.
- 18.** The one or more non-transitory computer-readable media of claim 14, wherein the one or more attributes comprises one or more of: an intent of the context usable by the retrieval engine to filter content items; an identification of the retrieval engine suitable for the context; and an identification of the content library suitable for the context.
- 19.** A system comprising: one or more processors; and one or more non-transitory computer-

readable media storing instructions that, when executed by the one or more processors, cause the one or more processors to: receive a context, wherein the context includes a natural language query from a user, one or more contextual factors about the user, and one or more contextual factors about a device used by the user; input, into a large language model, a prompt comprising the context and an instruction to produce a response about the context in an encoded output format; receive, from the large language model, the response generated based on the prompt, wherein the response comprises one or more encoded values corresponding to one or more attributes about the context; translate the one or more encoded values into a structured query; input the structured query into a retrieval engine to retrieve content items that correspond to the context from a content library; and output the content items to the user.

20. The system of claim 19, wherein translating the one or more encoded values into the structured query comprises: applying the one or more encoded values to structured query generation logic that produces structured queries conditioned on the one or more encoded values.
